**SE Lab 6: Code and Branch Coverage Analysis**

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## 1. Objective

To understand line and branch coverage using `pytest` and `coverage`, and write test cases to achieve >90% code and branch coverage for `pizza_order.py`.

## 3. Initial Baseline

The initial starter test file had only 2 tests.

Line Coverage: 41%

Branch Coverage: 34%

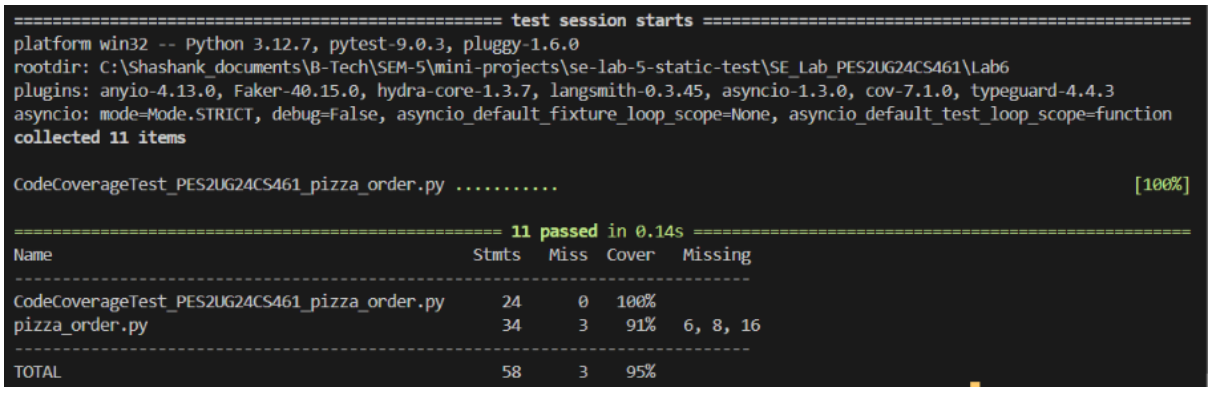
Many conditions like day discounts, delivery delays, and errors were untested.

## 4. Code Coverage Analysis

Added 9 test cases in CodeCoverageTest_PES2UG24CS461_pizza_order.py to test standard pizza orders, discounts, and delivery statuses.

Coverage Achieved 91% on pizza_order.py (95% overall).

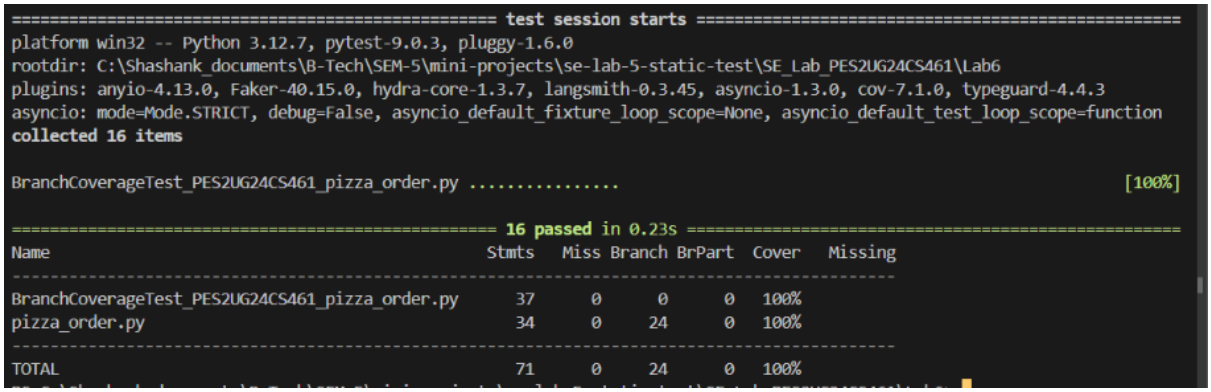
Missing lines: Lines 6, 8, and 16 (error cases).



### 5. Branch Coverage Analysis

Added test cases in BranchCoverageTest_PES2UG24CS461_pizza_order.py to test all decision branches and edge cases (wrong size, negative toppings, cheese burst on small, weekend vs weekday, delivery times).

Branch Coverage Achieved:100%  (24/24 branches covered, 0 missing).




### 6. Conclusion

Successfully improved the test coverage of pizza_order.py from 41% to 91% code coverage and finally to 100% branch coverage using targeted unit tests.